A Deep Quantum Convolutional Neural Network Based Facial Expression Recognition For Mental Health Analysis

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***Abstract*—The purpose of this work is to analyze how new technologies can enhance clinical practice while also examining the physical traits of emotional expressiveness of face expression in a number of psychiatric illnesses. Hence, in this work, an automatic facial expression recognition system has been proposed that analyzes static, sequential, or video facial images from medical healthcare data to detect emotions in people's facial regions. The proposed method has been implemented in five steps. The first step is image preprocessing, where a facial region of interest has been segmented from the input image. The second component includes a classical deep feature representation and the quantum part that involves successive sets of quantum convolutional layers followed by random quantum variational circuits for feature learning. Here, the proposed system has attained a faster training approach using the proposed quantum convolutional neural network approach that takes $O(\log(n))$ time. In contrast, the classical convolutional neural network models have $O(n^2)$ time. Additionally, some performance improvement techniques, such as image augmentation, fine-tuning, matrix normalization, and transfer learning methods, have been applied to the recognition system. Finally, the scores due to classical and quantum deep learning models are fused to improve the performance of the proposed method. Extensive experimentation with Karolinska-directed emotional faces (KDEF), Static Facial Expressions in the Wild (SFEW 2.0), and Facial Expression Recognition 2013 (FER-2013) benchmark databases and compared with other state-of-the-art methods that show the improvement of the proposed system.**

***Index Terms*—Facial expression, recognition, quantum, convolutional neural network, quanvolutional, quantum variational circuit, mental health conditions.**

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I. Introduction

TODAY'S community is plagued most frequently by mental problems. Mental health is a factor in determining a person's social, emotional, and psychological state. It influences one's thoughts, emotions, and approach to a circumstance. People can function and fulfil their potential in a productive manner with good mental health. In all stages of life, from childhood to adulthood, mental health is crucial. It has been demonstrated that mental problems, such as eating disorders, anxiety disorders, and depression, are exceedingly common and can affect a person's physical state. Numerous studies have highlighted the negative effects of social media use on mental health; thus, caution is still advised. Therefore, social media has provided a variety of opinion data about mental health issues. In most cases, the information is related to public feeling, while driving sentiments from those feelings are very tough. Even the baseline of sentiments can not be predicted about people's feelings [1]. Several techniques exist to derive sentiments from text, but the information from images and videos is more used in practice. As far as driving sentiments from people's feelings through images and videos, the public facial emotion recognition model comes forward first for the solution to the problem [2]. Facial expression plays an emergent role in our day-by-day communication and has an unmistakable indication of intellectual movement, affective state, personality, attention, self-aided driving, and an indication of thought or memory. It is a quick, effective, and powerful method of nonverbal communication to pass messages and convey emotional indication [3]. A facial expression recognition system (FERS) is considered to be a significant part of the Natural User Interface, where the human brain can easily recognize the emotional states of mind by looking at a glance at the muscular movement, contracted, and characteristics of faces. An automatic FERS is an open class problem where the face image $\mathcal{I}_{n\times n}$ is input. There are seven basic emotion classes, e.g., fear (AF), anger (AN), disgust (DI), happy (HA), neutral (NE), sad (SA), and surprise (SU), which is output. Fig. 1 shows a few samples of facial emotional expressions.

A comparative literature review for facial emotion recognition has been done in [4]. In [5], the few active patches from crucial action units and regions of interest of facial

Fig. 1. Examples of facial expression classes.

images have been employed. Hossain et al. [1] proposed FERS for Healthcare applications that can benefit patients, medical professionals, doctors, and medicine sellers. Ly et al. [6] have proposed a complex deep-learning technique for FERS by jointly fusing 2D and 3D facial features. Gu et al. [7] used the Gabor features-based FERS. Wu et al. [8] proposed a Discrete Beamlet transform regularization as a Blind Restoration model for infrared facial expression recognition problems. Zhi et al. [9] used a dynamic temporal 3DLeNets model for facial expression video analysis. The Quantum-Inspired Binary Gravitational Search Algorithm's feature selection method has been used to create an FERS in [10]. In [11], a quantum parameterized circuit-based quantum deep convolutional neural network (QDCNN) model for image recognition is examined. Recognizing facial expressions under varying yaw angles, Quantum-Inspired Particle Swarm Optimization has been demonstrated in [12]. Hence, there is no direct solution for the solution FERS using QCNN with its enhanced QCNN. In this work, a novel facial expression recognition system has been proposed to accept the discussed challenging issues. with the help of deep learning frameworks such as convolutional neural networks (CNN) and Quntum CNN, which provide beautiful results for facial expression recognition. Here, the contribution of our proposed work has been stated in the following:

(i) A quantum convolutional neural network (QCNN) model is proposed for the facial expression recognition system (FERS).

(ii) This quantum convolutional neural network (QCNN) model adopts a bilinear quadratic deformation cost function between the intermediate feature maps to establish an enhanced QCNN model (EQCNN).

(iii) The proposed EQCNN model increases the generalization and representation ability for multi-scale FERS under diversified image acquisition conditions.

(iv) The employed bi-linear quadratic deformation cost function reduces error and improves the recognition performance of FERS.

(v) The proposed system provides a faster Training approach using the proposed QCNN that takes $\mathbb{O}(\log(n))$ time, whereas the classical CNN takes $\mathbb{O}\left(n^2\right)$ time over the speedup of the recognition system.

The organization of this work is as follows: the proposed system's implementation has been discussed in Section II. The experimental results and discussion have been described in Section III. Finally, Section IV concludes this paper.

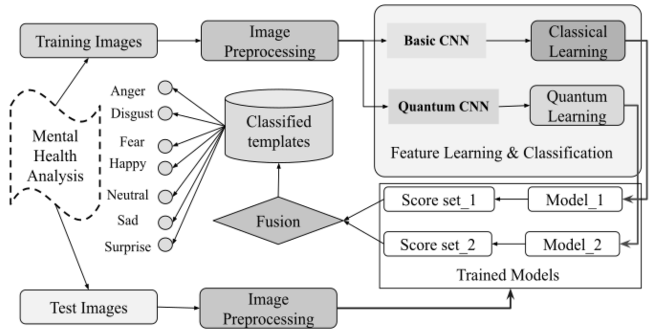

Fig. 2. Block diagram of the proposed FERS.

II. Proposed Methodology

Facial expression recognition, a crucial aspect of social cognition, is one of the diagnostic abnormalities of schizophrenia. Researchers have discovered that people with schizophrenia have difficulty identifying facial expressions of anger and sadness. People with schizophrenia are better at recognizing good emotions like Happiness than they are at recognizing negative emotions like despair and wrath. Such emotion-specific deficiency in schizophrenia patients is brought on by abnormal neuronal processing in brain areas that regulate the identification of unpleasant emotions in particular. An automatic facial expression recognition [1] from static, sequential, or video facial images has been proposed using a deep quantum convolutional neural network (QCNN) model for emotion recognition problems. Quantum Convolutional Neural Networks (QCNN) is the name given to the method that combines convolutional neural networks (CNN) and quantum computing. The drawback of convolutional neural networks is that they are less effective at learning when the size of the input or model is huge. Consequently, the introduction of quantum processing into CNN leads to the development of a more effective and superior technique that can be used to address challenging machine learning problems. Moreover, the fundamental components and architecture of conventional CNN are extended to quantum systems by quantum convolutional neural networks. The data size increases exponentially in direct proportion to the system size when a quantum physics problem stated in the many-body Hilbert space is translated to a classical computing environment, making it unsuitable for effective solutions. The issue can be solved by utilizing a CNN structure in a quantum computer because qubits can be used to express data in a quantum environment. The description of the proposed system has been demonstrated in Fig. 2.

The proposed method uses quantum computing in quantum convolutional layers that perform complex kernel operations in higher dimensional feature space to propose a new enhanced quantum convolutional neural network (EQCNN) model. It combines profound local discriminative features generated by convolutional neural networks (CNNs) and handcrafted features, including geometric shapes and appearance-based information, to enhance the proposed model's robustness and recognition performance. Instead of these, most deep learning frameworks use linear filters that generate good results but are not always optimal for real-life applications, especially for highly skewed multi-scale facial images ($\mathcal{I}_{n\times n\times r}$). This is because linear filtering has a limited response to its corresponding feature map; hence, to overcome these issues, quantum computing is embedded into the deep learning models. Here, the linear filters are replaced with quantum gates at each convolutional layer in this proposed network that establishes a self-training quantum neural network (QCNN) model.

The implementation of this work is divided into four segments: the first part involves face detection and pre-processing steps to extract and crop the facial region. The second part involves successive sets of quantum convolutional layers and pooling layers for encoding facial texture features into random quantum variational circuits (RQVC). Here, the convolutional layers use quantum filters to navigate input facial images through the quantum circuits. The third part includes deep local feature representation schemes using a set of convolutional layers, pooling layers, activation, and mini-batch operators. The CNN layers of this part learn more in-depth local features individually for training the model. The final part contains a classifier to perform the prediction of facial expressions. The model EQCNN derives features from the lack of abundant labelled and unlabelled highly skewed facial images [13] using quantum convolution filters & quantum circuits. The proposed quantum convolution neural networks model has used CNN and image augmentation methods to calculate handcraft and deep features. Here, the proposed model has increasingly been leveraged to learn discriminative representations of handcraft features [4]. We have also used the quantum transfer learning technique and the score level fusion.

A. Face Pre-Processing

This paper handles multi-scale facial images captured under unconstrained imaging conditions. The input facial images $\mathcal{I}_{m\times n\times r}$ have proceeded through the pre-processing steps to normalize the input images into the standard scale regarding size and color channels. The pre-processing steps aim to enhance the input facial expression region of interest (ROI) image and suppress the unwanted, inconsistent, and redundant noise in the image [1]. To extract the ROI from the input image $\mathcal{I}(n \times n \times 3)$, we have used enhanced the tree structure part models [14] to extract sixty-eight (68) and eighty-one (81) landmarks. We have used the best shape predictors, i.e., sixty-eight landmarks for high quality and eighty-one landmarks for low quality, to obtain the full face. The high-quality full-face landmark detection accuracy is improved by repositioning forehead landmark coordinates to a more accurate position. Then, we cropped the required facial region. The process of image preprocessing contributes to preserving the edge texture features through the image smoothing process. It minimizes intensity variations by substituting the average value for each texture feature in the image. The quanvolution filters are used then to eliminate noise, distortion, and redundant information from undesired image characteristics. The pre-processing tasks have been shown in the following Fig. 3.

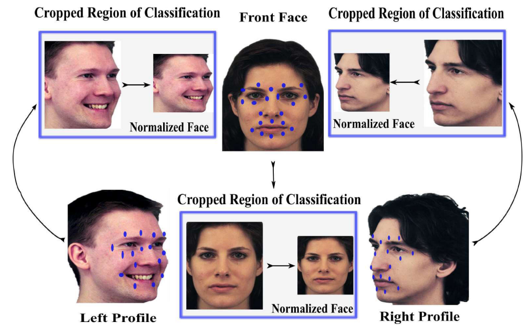

Fig. 3. Face pre-processing mechanism for proposed FERS.

B. Feature Representation

In affective computing, feature extraction is the most crucial job as it is related to dimensionality reduction using special feature map functions when the input image is too large to process by the model. Here, the proposed feature extraction scheme provides a semantic and robust representation using CNNs by extracting features from multi-scaled facial image $\mathcal{I}_{n\times n}$ to build a robust and accurate FERS. The CNN models extract informative and non-redundant texture features by employing successive operations like convolution, batch normalization, activation, and max-pooling by sliding a smaller kernel along the input image. At the same time, quantum computing can deal with higher dimensional feature space by estimating complex kernels in n-dimensional space. However, CNN still has many drawbacks, such as not extracting significant discriminative facial features and losing helpful information during the feature extraction phase, leading to low-performance recognition rates in multi-scale facial feature extraction and representation. This work proposes a method based on computational complexity and multi-scale feature fusion techniques using quantum convolution neural networks (QCNN) to overcome these problems. The quanvolution kernel of homogenous scales brings the receptive fields for various scales in image resolution.

C. Proposed Basic CNN Architecture

We have selected input image $\mathcal{F}_{n\times n\times r}$ and convolved with a set of filters known as kernels of size $(t_1 \times t_1)$, here $t_1 = 3$. Here, the mechanism of these hidden convolutional layers is known as feature function or mapping. The feature maps are stacked with fixed-size kernels to provide multiple filters on the input. We have employed (3×3) filter's size having stride one for each hidden convolution layer and Rectified Linear Unit $(ReLU)$ activation function for each hidden convolution layer. The computational complexity of the CNN_{Basic} model has been reduced by using (2×2) pooling layers, which reduces the output size from one layer to the next input hidden layers. Here, these images are downsampled to half of their original input size. The maps are flattened into one column to feed the pooled output from the stacked featured map to the final layer. The final layers comprise two fully dense connected layers with $\mathcal{N}$ number of hidden nodes each. These two flattened layers are regularized using the dropout regularization technique. Finally, the Softmax layer is followed

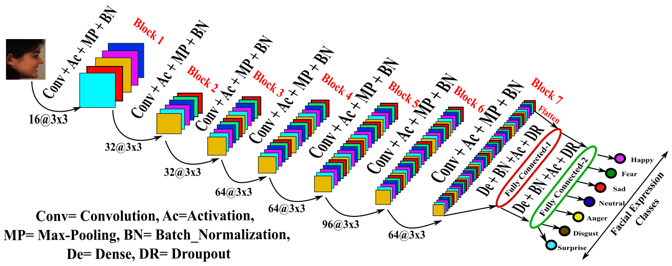

Fig. 4. Demonstration of the proposed basic *CNN* architecture taking $F_{n\times n\times 3}$ as input image.

TABLE I
DESCRIPTION OF PARAMETERS IN BASIC *CNN* WITH INPUT IMAGE SIZE AND OUTPUT SHAPE

Layers	OutputShape	ImageSize	Parameters
Block-1			
Conv2D(3x3)@16	(n,n,16)	(200, 200, 16)	((3x3x3)+1)x16=448
Maxpool2D(2x2)	$(n_1,n_1,16)$, $n_1 = n/2$	(100, 100, 16)	0
BatchNorm	$(n_1,n_1,16)$	(100,100, 16)	4x16=64
ActivationReLU	$(n_1,n_1,16)$	(100, 100, 16)	0
Dropout	$(n_1,n_1,16)$	(100, 100, 16)	0
Block-2			
Conv2D(3x3)@32	$(n_1,n_1,32)$	(100, 100, 32)	((3x3x16)+1)x32=4640
Maxpool2D(2x2)	$(n_2,n_2, 32)$, $n_2 = n_1/2$	(50,50,32)	0
BatchNorm	$(n_2,n_2,32)$	(50, 50, 32)	4x32=128
ActivationReLU	$(n_2,n_2,32)$	(50, 50, 32)	0
Dropout	$(n_2,n_2,32)$	(50,50,32)	0

Layers	OutputShape	ImageSize	Parameters	Layers	OutputShape	ImageSize	Parameters
Block-3				**Block-4**			
Conv2D (3x3)@32	$(n_2,n_2,32)$	(50,50,32)	((3x3x32) + 1)x32 =9248	Conv2D (3x3)@64	$(n_3,n_3,64)$	(25,25,64)	((3x3x32)+1) x64=18496
Maxpool2D (2x2)	$(n_3,n_3,32)$ $n_3 = n_2/2$	(25, 25, 32)	0	Maxpool2D (2x2)	$(n_4,n_4,64)$ $n_4 = n_3/2$	(12,12,64)	0
Batch Norm	$(n_3,n_3,32)$	(25,25,32)	4x32 = 128	Batch Norm	$(n_4,n_4,64)$	(12,12,64)	4x64=256
Activation ReLU	$(n_3,n_3,32)$	(25,25,32)	0	Activation ReLU	$(n_4,n_4,64)$	(12,12,64)	0
Dropout	$(n_3,n_3,32)$	(25, 25, 32)	0	Dropout	$(n_4,n_4,64)$	(12,12,64)	0
Block-5				**Block-6**			
Conv2D (3x3)@64	$(n_4,n_4,64)$	(12, 12, 64)	((3x3x64)+1) x 64=36928	Conv2D (3x3)@96	$(n_5,n_5,96)$	(6,6,96)	((3x3x64)+1) x96=55392
Maxpool2D (2x2)	$(n_5,n_5,64)$ $n_5 = n_4/2$	(6, 6, 64)	0	Maxpool2D (2x2)	$(n_6,n_6,96)$ $n_6 = n_5/2$	(3,3,96)	0
Batch Norm	$(n_5,n_5,64)$	(6, 6, 64)	4x64=256	Batch Norm	$(n_6,n_6,96)$	(3,3,96)	4x96=384
Activation ReLU	$(n_5,n_5,32)$	(6, 6, 64)	0	Activation ReLU	$(n_6,n_6,96)$	(3,3,96)	0
Dropout	$(n_5,n_5,64)$	(6, 6, 64)	0	Dropout	$(n_6,n_6,64)$	(3,3,96)	0

Layers	OutputShape	ImageSize	Parameters
Block-7			
Conv2D(3x3)@64	$(n_6,n_6,64)$	(3,3,64)	((3x3x96)+1)64=55360
Maxpool2D(2x2)	$(n_7,n_7,64)$, $n_7 = n_6/2$	(1,1,64)	0
BatchNorm	$(n_7,n_7,64)$	(1,1,64)	4x64=256
ActivationReLU˜	$(n_7,n_7,64)$	(1,1,64)	0
Dropout	$(n_7,n_7\ ,64)$	(1,1,64)	0
Layer	**Output Shape**	**Image Size**	**Parameter**
Flatten	$(1,\ n_7 \times n_7 \times 64)$	1, 64	0
Dense	(1, 256)	(1, 256)	(1 + 64)x 256 =16640
Batch Normalization	(1, 256)	(1, 256)	1024
Activation Relu	(1, 256)	(1, 256)	0
Dropout	(1, 256)	(1, 256)	0
Dense	(1, 256)	(1, 256)	(256+1)x256 =65792
Batch Normalization	(1, 256)	(1, 256)	1024
Activation Relu	(1, 256)	(1, 256)	0
Dropout	(1, 256)	(1, 256)	0
Dense	(1, 7)	(1, 7)	(256+1)x7 =1799
Total Parameters for The Input Image:			267975
Total Number of Trainable Parameters:			266215
Non-trainable params:			1760

by two fully connected layers, and the number of nodes in this layer is equal to the number of emotion expression classes. The detailed description of the basic CNN model and the employed input image size, generated parameters at each input-output hidden layer, convolution layer, max-pooling layers, batch normalization, activation, and dropout layers have been described in Table I and the architecture of basic CNN have been depicted in Fig. 4 respectively for better understanding and analysis.

D. Proposed QCNN Architecture

In classical computing, binary 'bits' represent a unit of information, i.e., "0" or "1" for information processing. In quantum machine learning (QML), bits are represented by "qubits" or quantum bits. A qubit is analogous to classical bits or quantum variants of the bit, representing the small unit of information. Qubits are used to process information faster than classical computing. Qubits could be in a superposition

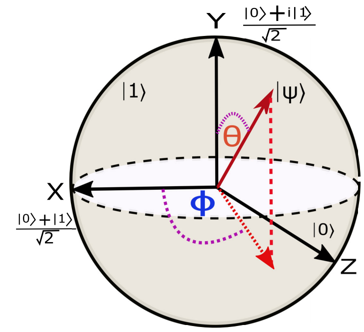

Fig. 5. Visually represent a qubit in superposition state $\mid \psi\rangle$.

states [15] i.e., the states represented by a qubit's state "$\mid 0\rangle$," or "$\mid 1\rangle$," or a linear combination of these two states at the same time called superposition states of the qubit's possible states shown in Fig. 5. The qubit represents state vectors in Hilbert space and is completely mutually exclusive. A qubit is represented by orthogonal vectors i.e. $\mid 0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\mid 1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ or $\mid \psi\rangle = \begin{pmatrix} \theta \\ \phi \end{pmatrix} = \theta \mid 0\rangle + \phi \mid 1\rangle$, where θ and ϕ are probability amplitude calculated by $\mid \theta^2 \mid + \mid \phi^2 \mid = 1$. They have the same restrictions in a quantum computing environment as normal bits in classical computing. They can store only a single piece of information and can only give us an output of $\mid 0\rangle$ or $\mid 1\rangle$ or $\mid \psi\rangle$. Qubit could rotate along the $\mathcal{X}$, $\mathcal{Y}$, and $\mathcal{Z}$ axis from any given point of time, and in 2π intervals, the qubit repeats the states. Therefore, a quantum computer with **n** qubits can have all $\mathbf{2^n}$ possibilities in superposition states. Quantum gates take qubits as input and perform a fixed quantum computation operation or parallel computation operation like performing rotation on a qubit, flipping a qubit state, or changing the state of one qubit based on the control qubit.

There are various types of quantum gates like single qubit gates or 2×2 unitary matrices, i.e., Pauli gates, two-qubits gates or 4×4 unitary matrices [16] i.e. Controlled NOT gate, and multiple-qubits gates or $2^n \times 2^n$ i.e. Toffoli gates [16]. Hence, $\mathcal{R}_i X(\theta_i)$ states that a single qubit gate rotates the i^{th} qubit along $\mathcal{X}$-axis by an angle θ_i. The probability of measuring the states $\mid \psi\rangle$ and $\mid x\rangle$ of the quantum circuit is $p(\mid x\rangle) =\mid \langle x \mid \psi\rangle \mid^2$. Here, $\langle x \mid$ represents row vectors called "bras," and $\mid \psi\rangle$ is used to represent column vectors known as "ket" in quantum computers. Since "$\mid$ *and*$\rangle$" and "$\langle$*and* $\mid$ "make bracket notation. In quantum computing, a unitary matrix satisfies the property: $\mathbb{U}\mathbb{U}^\tau = \mathbb{U}^\tau\mathbb{U} = \mathbb{I}$, where $\mathbb{U}^\tau$ is the conjugate transpose of a matrix $\mathbb{U}$ and $\mathbb{I}$ is the identity matrix. A quantum gate represents the unitary transformation of a unitary matrix [17].

A Quantum Convolution Neural Network (QCNN) uses the structure of CNN by staking a set of sequential quanvolutional layers, pooling operators, batch normalization, activation functions, and quantum kernels. A quanvolutional layer acts and behaves just like a convolutional layer. Here, the convolutional neural network (CNN) is a standard model for feature learning and image classification for processing facial images $\mathcal{I}(n \times n)$. It finds new deeply hidden features by maintaining correlation information with a linear combination between local surrounding texture features. The QNN model is based on the idea of a convolution layer where, instead of processing the full input image, a local convolution is applied to the deep local texture feature. From the input image, $\mathcal{I}_{(}n \times n)$, small local regions or action units (AUs) are selected and processed with the homogeneous quanvolutional kernel. The quanvolved deeper texture features obtained for each region are usually associated with different channels of a single output feature. The union of all the output features produces new orthogonal feature vectors representing deep local texture features for the facial image. These extracted features are mutually exclusive and can be further processed by adding successive layers. Here, we have extended the idea of classical convoluted computation into quantum computation and the context of quantum variational circuits (QVC). A small region from the input image $\mathcal{I}_{(}n \times n)$, for experimentation, a 2×2 square, is embedded into a random quantum variational circuit. This is achieved with parameterized rotations $\left(\mathcal{R}_y\left(\theta_i\right)\right)$ applied to the qubits initialization for encoding the input 2×2 square region, i.e., batches in the ground state.

A quantum computation associated with a unitary matrix, $\mathbb{U}$, is performed on the system. The unitary could be generated by a variational quantum circuit or, more simply, by a quantum random circuit (QRC). The quantum system finally measured quantum trainable parameters θ_i and obtained a list of classical expectation ρ_z as output in the $\mathcal{Z}$-basis vectors and mapped into four different channels of a single output texture feature vector. The qubit expectation values of the first hidden layer ρ_z are measured using the rule: $\rho_z = \sum_i^N z_i \mathcal{P}\left(z_i\right)$, are used as inputs to the next hidden layer. The results could be classically post-processed using the trained expectation ρ_z. Analogously to a basic classical convolution layer, each expectation ρ_z is mapped to a different channel like Ch.1, Ch.2, Ch.3, and Ch.4 as output features to down-sample the image and to locate distortion of the image, preserving the global shape. Iterating the similar procedure over the entire facial image tacking 2×2 square batches by scanning the full input image and moving starts from the top-left corner to the bottom-right corner, generating an output channel that will be structured as a multi-scale and multi-label image. Here, the mechanism performed by quantum computing is shown in Fig. 7. Quantum layers and pooling layers can follow quantum convolution. The pooling layers are used to reduce the feature map size to overcome the overfitting problem. The resulting features generated from quantum parts are fed into basic CNN layers for final classification tasks and optimum performance evaluation. The main difference concerning a classical convolution is that during feature encoding, the quantum circuit can generate highly complex kernels in higher dimensional feature space. The computation could be classically intractable, at least in principle.

In our proposed QCNN architecture, some dense layers are fully connected flattened layers in a quantum convolutional neural network [18]. In the dense layer, all inputs are connected to the output feature maps with specified weights and perform linear operations [1]. The linear operations use the following feature map function: $\zeta_l^b = \eta^b\left(\omega^\tau \zeta_{l-1}^b\right)$, where η^b is the activation function ReLU defined as $\mu\left(\sigma\right) = max\left(0, \sigma\right)$. The pooling layers, followed by ReLU activation functions, are used to reduce the computational power of the model by decreasing the spatial size of the quanvolved feature vectors and dimensions. Here, the fully connected nodes in dense layers are decreased to overcome the overfitting problem and perform better.

E. Proposed EQCNN Architecture

Quantum computing is an emergent and major transformative technology and an active research field. Creating quantum convolutional neural networks (QCNN) for facial expression recognition out of multi-scale input face images is quite complex. We aim to take advantage of the quantum convolutional layer, merge with CNN, and produce a new proposed FER model for highly skewed facial images. Each input is considered quantum data, and a quantum circuit replaces each texture feature. A quantum circuit comprises a series of quantum gates and ancilla qubits connected through the deeper parameterized quantum circuit (PQC) that performs unitary matrix $\mathbb{U}$ operation on qubits. Ancilla qubits are extra controlled memory bits used in quantum gate circuits to achieve specific objectives during quantum computation. The advantages of using quantum circuits under a quantum computation environment are that it takes few quantum bits and depth deeper circuits. The advantage of using deeper circuits is to make our proposed model more robust and resilient against quantum noise. The quantum circuit is divided into three steps: the quantum data encoding part, the random variational circuit consisting of successive quantum convolutional filters, pooling layers, and the encoder or measurement part. The quantum data encoding parts are stable for specific QCNN structures and transform $\bar{\mathcal{N}}$ classical features into $\bar{\mathcal{N}}$ quantum data, i.e., into $\bar{\mathcal{N}}$- qubits. While the quantum filters, batch normalization, and pooling layers use parameterized ancilla qubits and quantum gates in the random quantum variational circuit.

There are successive quantum convolutional layers, and in each layer, multiple quantum filters are employed to the input feature map to generate new data. Here, the filters applied at i^{th}-layer are represented by f_i. At each hidden layer, the quantum filters perform operations on the same two-qubit ansatz to the nearest neighbor qubits in a translationally invariant method. The pooling layer reduces the feature map size and avoids overfitting problems. The measurement phase or decoding part transforms the final quantum data into classical form. In our proposed Enhanced Quantum Convolutional Neural Network (EQCNN) model, the pooling operations used in quanvolutional layers are illustrated as a controlled ancilla unitary transformation $\mathbb{U}$. A small 2×2 square region or patches from the input facial image $\mathcal{I}_{n\times n}$ is embedded into a quantum circuit. It is called a variational quantum circuit with a trainable quantum parameter, i.e., neuron or node α_i, and a scalar value called a weight, i.e., ω_i associated with each edge connected between two transformation layers. Neurons of successive layers are connected through trainable weights. The Variational Quantum Classifier (VQC) is considered a black box. The gradient of this black box concerning its quantum parameters can be calculated in Eq.1.

$$\nabla_\theta Quan\_Cir\ (\theta_i) = Quan\_Cir\ (\theta_i + \phi) - Quan\_Cir\ (\theta_i - \phi) \quad (1)$$

where θ is the quantum parameters, and ϕ is the macroscopic shift. Here, PennyLane Tensorflow packages are used to evaluate gradients of quantum variational circuits at each transformational layer and train the quantum kernel [19]. The proposed EQCNN is a custom class of quantum convolutional neural networks [19] for feature pre-processing [20]. Quantum computing uses Hilbert space to represent convolved orthogonal feature vectors. Hence, it is possible to construct complex kernel function $\mathcal{F}_x(|\ \psi_\alpha\rangle)$, which is suitable to fit complex non-linear data sets. We have constructed an EQCNN with VQC and Quan2D class for feature extraction and representation with successive convolutional layers followed by activation function ReLU, pooling layers, and batch normalization. The convolutional layers contain two rows of two-qubit unitaries, which act on alternating pairs of two neighboring qubits composed of variational parameters. Our custom quantum encoding layer relies on a feature map to encode classical data using parameterized rotations, followed by a fully connected circuit that maps classical input to qubit states. The pooling layer contains measurements on half of the qubits, with the outcome of the measurement controlling unitary acting on the neighboring qubits. The pooling operation of two qubits maps the 2-qubit Hilbert space to a 1-qubit Hilbert space. A convolutional layer follows each pooling layer until the final Hilbert space of the system. Each custom layer maps the quantum superposition states to classical data at the decoding layer. Hence, the classification tasks for FER have been solved using a variational quantum convolution neural network classifier. The architecture of our proposed EQCNN has been depicted in Fig. 6, where each R_i is the rotation with angle θ_j. The detailed descriptions of the proposed model architectures, along with employed input-output parameters at each hidden transformations layer, the generated feature vector shapes of the quanvoluted images, the input feature map shapes, and parameters produced at each layer, are shown in Table II, respectively, for simplicity, generalization and better understanding about the models.

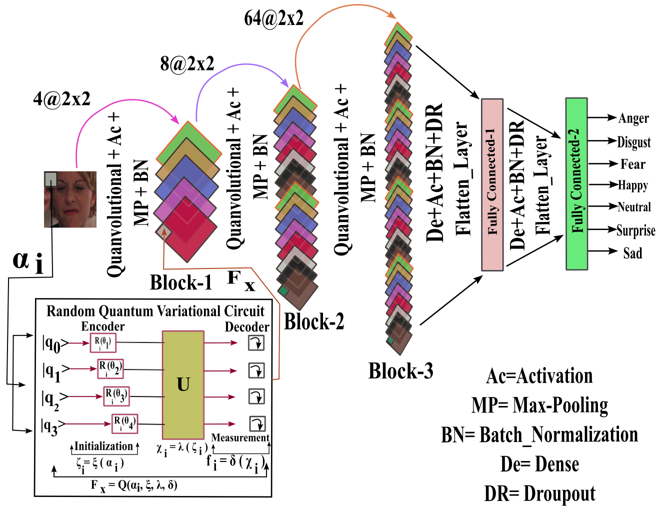

Fig. 6. Proposed $EQCNN_{n\times n}$ architecture for Facial Expression Recognition and Classification for input image $I_{n\times n}$.

TABLE II
PROPOSED EQCNN ARCHITECTURE FOR THE INPUT SIZE 120 × 120 WITH LAYERS, INPUT, OUTPUT SHAPES, AND PARAMETERS GENERATED

Layer	Input Shape	Output Shape	Image Size	Parameters
Block-1				
Qconv1 (QConv)	(120, 120) (n,n,4)	(n_1, n_1, 4) n_1 = (n-1)	(119, 119, 4)	24
MaxPooling 2D	(119, 119, 4) (n_1, n_1, 4)	(n_2, n_2, 4) n_2 = (n/2) - 1	59, 59, 4	0
Batch Normalization	(59, 59, 4) (n_2, n_2, 4)	(n_2, n_2, 4)	(59, 59, 4)	(4 x 4) = 16
Block-2				
Qconv2 (QConv)	(59, 59, 4) (n_2, n_2, 4)	(n_3, n_3, 8) n_3 = (n_2 - 1)	(58, 58, 8)	192
Max Polling_2D_1	(58, 58, 8) (n_3, n_3, 8)	(n_4, n_4, 8) n_4 = (n_3 / 2)	(29, 29, 8)	0
Batch Normalization_1	(29, 29, 8) (n_4, n_4, 8)	(n_4, n_4, 8)	(29, 29, 8)	(4 x 8) = 32
Block-3				
Qconv3 (QConv)	(28, 28, 64) (n-5, n_5, 64) n_5 = (n_4 - 1)	(n-5, n_5, 64)	(28, 28, 64)	3072
Max Pooling_2D_2	(14, 14, 64) (n_6, n_6, 64) n_6 = (n_5 / 2)	(n_6, n_6, 64)	(14, 14, 64)	0
Batch Normalization_2	(14, 14, 64) (n_6, n_6, 64)	(n_6, n_6, 64)	(14, 14, 64)	(4 x 64) = 256
Flatten	(14, 14, 64)	n_6 x n_6 x 64	(14 x 14 x 64) = 12544	0
Dense		(1, 1, 16)	(1, 1, 16)	(12544 x 16) = 200720
Batch Normalization_3		(1, 1, 16)	(1, 1, 16)	(4 x 16) =64
Dropout		(1, 1, 16)	(1, 1, 16)	0
Dense		(1, 1, 7)	(1, 1, 7)	(16 + 1) x 7 = 119
Total Parameters			204,495	
Trainable Parameters			204311	
Non-trainable Parameters				184

F. Computational Complexity Analysis in EQCNN

This paperwork resolves to incorporate an enhanced quanvolutional filter or kernel into classical basic CNN instead of a linear filter to identify and classify facial expression recognition effectively. The impact of CNN architecture on classification performance, including the number of filters and kernel size and different feature learning techniques, are explored in [1] and [2]. Here, quantum computing takes advantage of the entanglement of the quantum convolutional layers, i.e., the correlation between qubits to encode facial images that exploit interdependence between qubits computed from AUs and attributes or face regions of interest. The EQCNN uses automatic differentiation of quantum circuits for image encoding and optimization, potentially offering a more accurate result. Hence, the EQCNN has higher accuracy than the basic convolutional model based on finite differentiation. Here, the quantum kernel computation in quanvolutional layers is shown in Fig. 7.

G. Transfer Learning Technique in EQCNN

Here, we have extended the concept behind the classical transfer learning technique to the emergent context of hybrid quanvolutional transfer learning techniques. The quantum transfer learning techniques comprise classical elements, quantum circuits, qubit gates, quantum computing environment, quantum superposition state, entanglement, and quantum

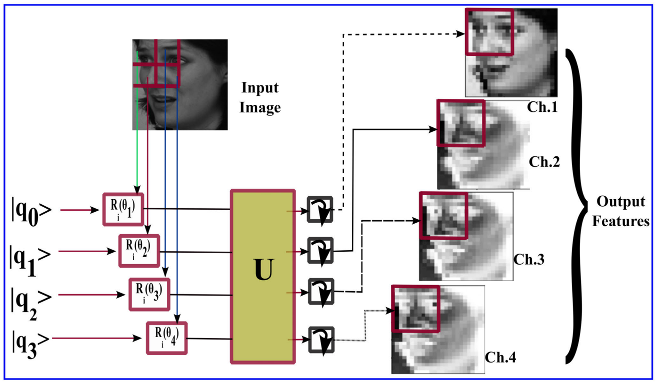

Fig. 7. Effect of the quantum convolution layer on a single batch of the input image in proposed EQCNN.

parallel computation operators [21]. Here, quantum transfer learning techniques are applied to validate the effectiveness of our proposed models and are effective, especially for texture synthesis, object detection, fine-grained texture recognition, visual question-answering, and image classification tasks. The principle concept behind quantum transfer learning is that low-level features learned from a model designed for a problem are used for the newly related problems as a generic model [22]. Here, standard benchmark facial images are first trained in the proposed enhanced quantum convolutional neural network (EQCNN) model. Then, the model was fine-tuned with a few shot pre-trained model from the target domain, where some input or output weights of hidden layers are frozen [23]. In classical computation, it is required $\mathcal{O}\left(n^2\right)$ operations for an $n \times n$ sized filter. In contrast, in the quantum computing environment, it is required $\mathcal{O}\left(\log\left(n\right)\right)$ operations using qubits parallelism. Hence, the feature learning and representation evaluation time is reduced in quantum parallel computation compared to classical computing. The potential advantage of QCNN lies in its computational efficiency in manipulating 2^n quantum states with a small number of qubit gates and quantum circuits [24]. We have presented a proof-of-concept of hybrid quantum transfer learning for facial image and quantum state classification by the random quantum variational circuit in Fig. III-C.

In our proposed work, dense layers are fully connected flattened layers in an EQCNN [18]. In the dense layer, all inputs are connected to the output feature maps with specified weights and perform linear operations [1]. The linear operations performed by the dense layer have used the following feature map function: $\zeta_l^b = \eta^b\left(\omega^\tau \zeta_{l-1}^b\right)$, where η^b is the activation function ReLU defined as $\mu\left(\sigma\right) = \max\left(0, \sigma\right)$. The pooling layers, followed by ReLU activation functions, are used to reduce the computational power of the model by decreasing the spatial size of the quanvolved feature vectors and dimensions. Here, the fully connected nodes in dense layers are decreased to overcome the over-fitting problem and perform better. Here, image augmentation techniques have been employed at training time to improve the ability to generalize the model and reduce overfitting. Here, the image data augmentation techniques using [25], each training image is flipped randomly, horizontally, and vertically and also adopted transformation techniques like unsharp filtering,

TABLE III
DESCRIPTION OF EMPLOYED DATABASES

Database	# of Class	# of Training Samples	# of Testing Samples	Remarks
KDEF	7	1210	1213	Spontaneous Posed Unconstrained
SFEW 2.0	7	773	653	Controlled Spontaneous Unconstrained
FER2013	7	28709	7178	Controlled Wild Constrained

Gaussian blur, image scaling, Gaussian noise, bilateral filtering, image rotation and translation, image filling, contrast normalization, image zooming, and shearing.

H. Implication of Fusion Techniques

Here, to enhance the recognition system's performance, the fusion techniques are applied at the match score level to get a final recognition decision. Two different fusion approaches can be used: feature-level fusion and score-level fusion. In this paper, the score level fusion techniques have been applied to the scores generated by $EQCNN_{60\times60}$ and $EQCNN_{120\times120}$ models for test images. To achieve higher performance, the classification scores obtained due to $EQCNN_{60\times60}$ ($\mathbb{S}_{EQCNN_{60\times60}} = \left(\alpha_1^1, \alpha_2^1, \cdots, \alpha_7^1\right)$) and $EQCNN_{120\times120}$ ($\mathbb{S}_{EQCNN_{120\times120}} = \left(\beta_1^2, \beta_2^2, \cdots, \beta_7^2\right)$) undergo the Product-Rule and Sum-Rule score fusion techniques. These techniques are defined as follows: (i) $\max_{i\neq j,\ k=\{1,\cdots,7\}}\{\alpha_k^i \times \beta_k^j\}$ (Product-rule); (ii) $\max_{i\neq j,\ k=\{1,\cdots,7\}}\{\alpha_k^i + \beta_k^j\}$ (Sum-rule).

III. EXPERIMENTAL RESULTS

For the experimental purpose, KDEF (Karolinska-directed emotional faces) [26], Static Facial Expressions in the Wild 2.0 (SFEW 2.0) [27], and FER2013 [28] benchmark databases are employed to validate the stability and effectiveness of the proposed method. KDEF database contains 70 different individuals and is composed of 1210 images for the training set and 1213 for the test set with seven facial expression classes, i.e., afraid, anger, disgust, happiness, neutral, sadness, and surprise [3] having pose variants with five different angles, illumination conditions, ethnicity, and geographical locations. SFEW 2.0 database was created from an acted facial expression video database in the wild, which is composed of 1426 images, divided into a training set of 773 images and a test set of 653 images with seven types of facial expression classes. The FER2013 database is composed of 30,000 facial RGB images of different expressions, which are divided into training and testing sets with seven categories of facial expressions: annoyance (AN), disgust (DI), anxiety (FE), gladness (HA), sadness (SA), surprise (SU), and neutral(NE). Fig. 8 shows some image samples of the KDEF, SFEW 2.0, and FER2013 databases, and Table III summarizes the description of these databases.

A. Result and Discussion

This section describes the experimental setup for the proposed FERS. This system has been implemented using

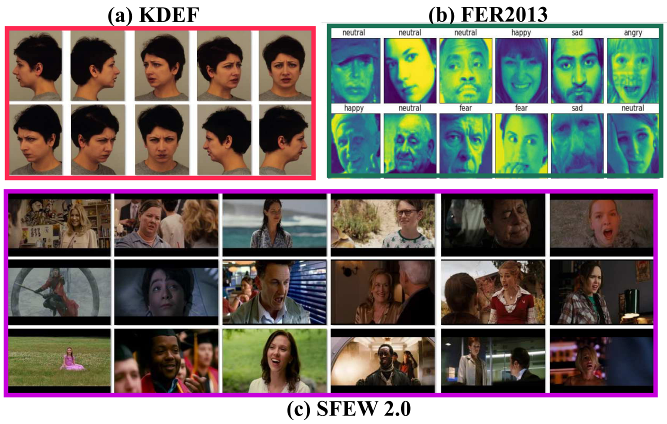

Fig. 8. Image samples of employed databases.

Tensorflow 2.9.1 version, Python 3.9 version, CUDA Version 11.2, Keras 2.4.3 version, and NVIDIA-SMI 460.79 Driver Version in Ubuntu 20.04.4 LTS in Windows 10 Pro 64-bit, 3.30GHz(8 CPU) Processor, Intel(R) Core(TM)-i7-9700 CPU, 8 GB NVIDIA GeForce RTX 2070 SUPER XLA GPU device with 16 GB RAM. Here, both grey-scaled and RGB-colored images are used during feature computation. During image pre-processing, a facial image is input image $\mathcal{I}_{m\times n\times 3}$ and then, the detected face region is normalized to a standard range size $\mathcal{F}_{n\times n\times 3} \in \mathbb{R}^{60\times 60\times 3}$ and $\mathcal{F}_{n\times n\times 3} \in \mathbb{R}^{120\times 120\times 3}$.

The proposed Enhanced quantum convolutional neural network (EQCNN) architecture uses quanvolution filters instead of linear filters in convolution layers. This EQCNN architecture is trained so that it uses a set of quanvolution layers with complex quantum kernels. Then, max-pooling mathematical operations after convolving the input facial image $\mathcal{F}_{(n\times n)}$ that down-samples each feature map to half its size. Using max-pooling layers suppresses noise and removes redundant features during spontaneous multi-scale facial images [1]. It helps to capture essential structural features of the represented images without being bogged down by the fine details. Some facial image databases are highly skewed and imbalanced, so batch normalization is used to prevent skews at a particular point and increase the computation speed. Here, batch normalization is employed after each quanvolutional layer to normalize the inputs from the previous layer at each batch [2]. The quantum convolutional neural network (QCNN) uses a complex quantum kernel. It is important to focus on two parameters, i.e., epoch and batches, during training time. Both of these two factors have a great impact on the proposed EQCNN architecture during learning features from the set of skewed facial images. Hence, we have established a trade-off between epochs and batches shown in Fig. 10 and 9. Initially, we built a $EQCNN_{60\times 60}$ model with image size 60×60 for experimentation. Next, we have up-sampled the facial images into 120×120 and constructed another model $EQCNN_{120\times 120}$. Here, two models are considered to train the spontaneous resolution and multi-pose facial images in the wild to introduce deeper local texture patterns at a higher abstraction level. The system could provide deeper information during encoding at each input variational quantum circuit. The performance demonstration with the trade-off between epochs and batches is shown in Fig. 9. From this figure,

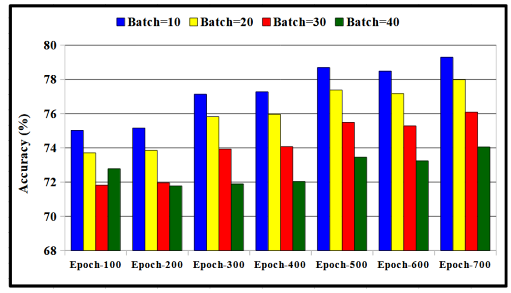

Fig. 9. Effect of varying batch-sizes with varying epochs using $EQCNN_{60\times 60}$ architecture on the proposed FERS using KDEF databases.

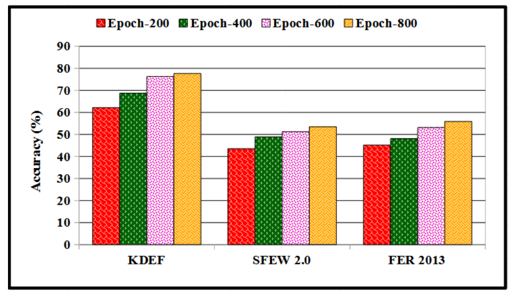

Fig. 10. Effect on the performance of the proposed FERS while keeping the batch fixed with varying epochs using $EQCNN_{60\times 60}$ architecture for the KDEF, SFEW 2.0, and FER 2013 databases.

It is observed that the performance improves gradually with increasing epochs for small batch sizes. The proposed model achieves superior results at higher epochs with small batch sizes, nearly between 10-20.

Fig. 10 is a complete representation of the consequence examination carried out by EQCNN, using the standard benchmark databases. Compared to the results demonstrated that the EQCNN model offered superior performance. It is also important to observe that as the total number of epochs grows, the accuracy will eventually increase. So, inspired by the performance reported in Fig. 10 and 9, the batch size is 10 and the number of epochs is 700 for further experiment.

B. Performance Using EQCNN and Basic CNN With the Effect of Data Augmentation

The performance of the proposed basic CNN architecture and EQCNN architecture has been reported in Table IV using 10 batch sizes. Here the effectiveness of employed data augmentation techniques has also been shown. It has been observed that the performance of the proposed FERS increases in both CNN and EQCNN architectures due to data augmentation. However, EQCNN architecture has achieved better performance with data augmentation, so these techniques are also adopted in the subsequent experiments.

C. Performance Due to Trade-off Between Quantum Transfer Learning and Basic CNN Transfer Learning

In this work, the quantum transfer learning techniques (QTL) are performed in the same approaches as classical CNN transfer learning techniques. So, here (i) we have newly trained $EQCNN_{60\times 60}$, and $EQCNN_{120\times 120}$ models corresponding to each image size, in such a way that the $EQCNN_{60\times 60}$ architecture is trained first, and then, with the progressive

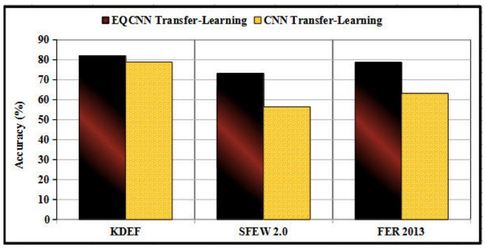

Fig. 11. A trade-off between Quantum Transfer Learning and Classical Basic CNN Transfer Learning.

TABLE IV
PERFORMANCE (IN ACC. (%)) OF FERS USING BASIC CNN AND $EQCNN_{60\times60}$ ARCHITECTURE

Database	Data Aug.	No Data Aug.	Data Aug.	No Data Aug.
	Basic CNN		EQCNN	
KDEF	76.45	73.24	78.54	76.51
SFEW 2.0	55.43	51.98	68.19	64.63
FER2013	62.78	61.30	75.06	74.55

TABLE V
PROPOSED FERS'S PERFORMANCE USING $EQCNN_{120\times120}$ QCNN ARCHITECTURE

Database	Precision	Recall	Acc.	F-Score
KDEF	63.52	69.19	81.95	74.50
SFEW 2.0	63.68	64.82	73.55	65.13
FER-2013	56.34	61.55	78.70	61.37

image sizing 120×120, the $CNN_{120\times120}$ architecture is retrained such that only upper layers of $EQCNN_{120\times120}$ model is trained, while the lower layers of $EQCNN_{120\times120}$ remains untrained. So, in the classical quantum transfer learning of EQCNN architecture, both (i) freezing some layers and (ii) whole model trained approaches have been employed, and for these approaches, the performance of FERS is affected. Here, the trade-off between quantum transfer learning and basic classical CNN transfer learning has been shown in Fig. 11 regarding performance and accuracy.

From this figure, it has been observed that the performance of our proposed EQCNN is better due to quantum transfer learning than classical CNN transfer learning for some highly skewed and high variance multi-scale facial images. Meanwhile, for some low-variance and highly skewed facial databases, the proposed EQCNN transfer learning technique is more suitable than classical CNN transfer learning. The overall performance of the proposed FERS is reported in Table V, which reveals the EQCNN model's effective performance on each database with diversified evaluation metric systems like "valence" and "arousal".

D. Performance Due to Fusion

The fused performance is shown in Table VI, which shows the performance is better for our proposed model in Fig. 6 for the fusion technique. To compare our results with diversified existing state-of-the-art methods on KDEF, SFEW 2.0, and FER2013, we have considered the performance of recognition systems under homogeneous training and testing protocol. Here, Table VII, Table VIII and Table IX show the performance of our proposed system with the other competing state-of-the-art methods for KDEF, SFEW 2.0, and FER2013 databases respectively. The feature vectors from the respective papers are implemented, and the performance is obtained in the same experimental setup used by the proposed system. The performance comparison reported in Table VII, VIII, and IX show the superiority of the proposed system. For better understanding and clarity, the confusion matrix corresponding to each employed database has been demonstrated in Fig.12.

Hence, the analysis of the time complexity of the proposed system with other competing CNN models has been provided in Table X.

TABLE VI
EFFECT OF SCORE-LEVEL FUSION ON PROPOSED FERS

Method	KDEF	SFEW 2.0	FER2013
$EQCNN_{60\times60}$	78.95	70.25	75.48
$EQCNN_{120\times120}$	81.85	72.52	78.70
Sum-Rule	81.53	72.95	79.51
Product-Rule	**81.95**	**73.55**	**79.95**

TABLE VII
PERFORMANCE COMPARISON USING KDEF DATABASE

Method	Acc.(%)	Remarks (Images, Expressions)
Vgg16 [29]	66.15	(980, 7)
ResNet50 [3]	74.05	(980, 7)
Zavare [26]	73.35	(980, 7)
Inception-v3 [30]	76.23	(980, 7)
Rao [31]	75.52	(720, 6)
Proposed	**81.95**	(980, 7)

TABLE VIII
PERFORMANCE COMPARISON USING SFEW 2.0

Method	Acc. (%)
Vgg16 [29]	26.01
ResNet50 [3]	26.98
Inception-v3 [30]	31.56
Liu et al. [32]	28.78
Proposed	**73.55**

TABLE IX
PERFORMANCE COMPARISON USING FER 2013

Method	Acc. (%)
SURF [33]	60.70
MTCNN [34]	65.03
Compact CNN [35]	62.44
HoloNet [36]	61.86
Proposed	**79.95**

TABLE X
COMPARISON OF TIME COMPLEXITY OF THE PROPOSED EQCNN AND OTHER COMPETING CNN ARCHITECTURES

Architecture	EQCNN	QCNN	CNN
Parameters	41,679	115,605	304,839
Time	$O(\log(n))$	$O(n^k \times j$	$O(n^k \times m \times j)$

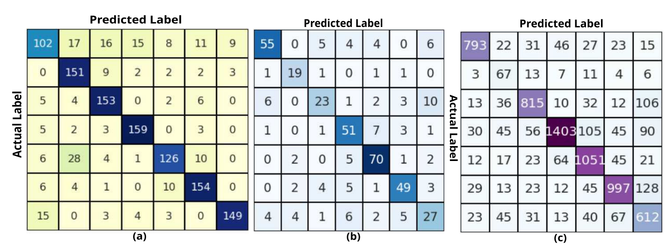

Fig. 12. Confusion matrix corresponds to (a) KDEF, (b) SFEW 2.0, (c) FER2013 database for the proposed system.

Where, k internal hidden layers, m number of epoch, j represents the number of iterations at each internal layer.

IV. Conclusion

Numerous research has revealed that physiological and behavioural signs, such as brain activity, galvanic skin response, eye contact, voice, and facial movements, are different in people with mental diseases. It can be difficult to recognize the early signs of any other mental disease. The proposed system offers a hybrid architecture that triggers an emotion sequence depending on facial expressions. It can also benefit from the ability to include systems for suspect detection and boosting intellect in people with brain development issues. Here, the proposed system is based on an enhanced quantum convolutional neural network; instead of linear filters, quantum convolutional filters are used. The implementation of this system is in four steps: (i) Image pre-processing, where from a body silhouette image, the face region has been extracted; (ii) advanced quantum convolution kernel and quantum computing techniques are used for feature extraction and representation; (iii) convolved orthogonal feature vectors obtained from the quantum part are fed into the convolutional neural network model for better feature representation and performance; (iv) scores due to convolutional neural networks and enhanced quantum convolutional neural networks are fused to obtain better performance. The performance of the proposed model has been tested and compared with the state-of-the-art methods using KDEF, SFEW 2.0, and FER2013 databases, which show the proposed model outperforms other competing methods.

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